Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence

Andrew Walther[1], Ashkan Habib[1], Ross Joseph Simpson, Jr.[2], and Feng-Chang Lin[1]

[1]Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA
[2]Division of Cardiology, University of North Carolina School of Medicine, Chapel Hill, NC, USA
Correspondence: Feng-Chang Lin, Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA. Email: flin33@email.unc.edu.

Abstract

Background/Aims: Spatial spillover can make treatment-effect estimation depend on how clusters are assigned to treatment. We compared average estimation error and allocation-specific downside across eight assignment strategies and simple random sampling (SRS), including a completed application for planning a sudden unexpected death (SUD) prevention education pilot.
Methods: A 100-cluster grid study varied incidence heterogeneity, spatial dependence, spillover magnitude, spillover regime and treatment-effect size. Maximum likelihood spatial autoregressive models estimated a known structural intervention effect. A focused nested pilot estimated conditional MSE across allocations. The application held observed 2018–2021 SUD incidence fixed for 58 North Carolina community-college service-area clusters and simulated continuous education outcomes. Primary application incidence guided assignment without entering the outcome baseline. Queen weights were primary; rook, hypothetical baseline/adjustment and regional-summary sensitivities were assessed.
Results: On the grid at effect size one, incidence-guided saturation had control-only mean MSE 0.1355 versus SRS 0.2146, while SRS remained competitive under both-arms spillover. In the completed application, control-only mean MSE was 27.4% lower for plain saturation and 21.1–29.9% lower for guided saturation than SRS, depending on year. Refined allocation-tail estimates also favored saturation under control-only spillover, with uncertain worst-decile membership. Under both arms, SRS and several randomized strategies had similar mean accuracy. Application yearly mean coverage was approximately 92.6–93.6%, below nominal 95%. All prespecified application Monte Carlo precision gates passed, with no warnings, aliases, boundary fits or estimation failures.
Conclusions: Saturation merits consideration when spillover primarily reaches controls; SRS remains competitive when both arms receive spillover. Incidence-guided saturation performed well on both average error and allocation downside under control-only spillover. Recommendations require attention to model assumptions, treatment budgets, interval undercoverage and finite allocation-tail precision. The application evaluates trial design, not observed intervention effectiveness.

Keywords: cluster randomized trial; spatial spillover; treatment allocation; simulation; mean squared error; sudden unexpected death.

Reading preview only. Final author approval remains pending.

Introduction

Cluster randomized trials assign groups rather than individuals when interventions operate through clinics, schools or geographic areas.[1,2] Spatial interference challenges the assumption that a cluster's outcome depends only on its own treatment: adjacent populations can share services, information or environmental exposures.[3,4] An allocation may therefore provide substantially different information about a treatment effect than another allocation with the same number of treated clusters.

Design-stage methods include covariate-constrained randomization, randomized saturation and independent-set or spatial cluster designs.[5–8] They address different objectives and constraints. Balancing measured incidence does not ensure treatment and spillover are distinguishable; maximal interspersion can itself make those covariates dependent. Appropriate spatial analysis remains necessary alongside allocation design.

In previous work, we compared simple random assignment and block-stratified sampling on small grids.[9] Its practical argument considered reasonably good estimation together with limiting poor-allocation downside in the studied settings. We extend that question to heterogeneous incidence, a larger grid, additional strategies and an irregular observed geography. We include both SRS and the strict alternating allocation called Checkerboard, while evaluating mean MSE and the distribution of conditional MSE across assignments. We do not directly compare numerical results between studies.

Our application supports planning an education intervention for healthcare professionals involved in SUD prevention. Historical incidence can identify areas of need without implying that education ability, or the intervention's effect on it, depends on incidence. We consequently separate incidence-guided allocation from the simulated education baseline. A trial may realize only one assignment, making both expected accuracy and poor-allocation downside relevant.

Incidence-informed assignment also raises a practical distinction. A trial may direct resources toward areas with greater historical disease burden, balance that burden between arms, or use it to determine local treatment saturation. These strategies express different priorities even when their overall treatment budgets are similar. They need not provide the same information for separating treatment from neighboring exposure. Our objective was to evaluate their estimation consequences under specified spatial models, including the possibility that the design with the lowest average error offers little additional protection against an unfavorable realized allocation.

Methods

Grid model and assignments

We used a 10×10 grid with row-standardized weights W. Queen neighbors share an edge or corner; rook neighbors share an edge. Queen was primary, with rook assessed

separately. Outcomes followed

$$Y = (I - \rho W)^{-1}[\tau Z + S(Z) + \beta X + \epsilon], \quad \epsilon \sim N(0, I), \tag{1}$$

where Z is treatment and τ its structural coefficient before spatial propagation. It is not an average total effect after applying the spatial multiplier. We set $\beta = 1$ in the grid study. Spillover was $S(Z) = \gamma WZ$ for both arms or $S(Z) = \gamma(1 - Z) \circ WZ$ for controls only, with elementwise product $\circ$. The parameter grid crossed $\rho = 0, 0.01, 0.2, 0.5$, $\gamma = 0.5, 0.6, 0.7, 0.8$ and $\tau = 0.8, 1, 1.5, 2, 3$.

Five incidence configurations comprised independent Uniform$(0, 1)$ incidence, spatial incidence at two dependence strengths and Poisson incidence at the same two strengths. Spatial incidence used a queen SAR filter followed by a normal distribution-function transformation. Poisson counts used spatial log-rates with base rate 35 per 100,000 and equal populations of 100,000; observed rates were rank-scaled to $(r_i - 0.5)/100$, using average ranks for ties. These are general simulation settings, not calibration to the application counts. Each configuration's ten independent surfaces were shared across parameters. For surface k, assignment, outcome generation and fitted incidence all used X_k.

The eight strategies were: Checkerboard; treatment of the highest-incidence half; plain saturation quadrants; Isolation Buffer, a greedy random maximal independent set; randomization of two clusters in every 2×2 block; balancing by incidence quartiles; balancing by incidence halves; and incidence-guided saturation quadrants. Both saturation strategies assigned 20/40/60/80% treatment within four quadrants; plain saturation randomized their order and guided saturation ordered them by mean incidence. All treated 50 clusters except buffer, whose budget depended on the graph. SRS completely randomized 50 of 100. Table 1 summarizes the rules and their geographic adaptations; supplementary material gives full specifications. Checkerboard was deterministic; incidence ties were broken randomly for each assignment in ranking-based designs. High Incidence Focus had singleton support when no tie affected selection.

The strategies vary both incidence balance and spatial exposure. Quartile and half stratification distribute treatment within incidence groups; High Incidence Focus instead aligns treatment with incidence. Spatial blocking imposes local treatment balance, while saturation permits intentionally different local proportions. Buffer prevents treated neighbors, and Checkerboard imposes strong interspersion. These restrictions can affect the relationship between treatment and spillover covariates, beyond the number treated. The detailed grid analysis retained six designs after comparing the two related variants, plain versus guided saturation and halves versus quartiles. We nevertheless include all eight strategies and SRS in the full benchmark and application: near performance of related designs does not establish equivalence or justify ignoring their practical differences.

The nine-design benchmark contained 14,400 scenarios per estimator, each with ten surfaces and 25 assignment/outcome pairs per surface, giving 250 independently noised fits. Designs within a block shared surfaces and outcome noise; treatment-effect levels reused assignment/noise draws. The SRS extension preserved the original eight-design outputs. Scenario metrics were MSE, bias and 95% Wald coverage. Surface-level Monte Carlo SEs and paired comparisons used independent configuration-by-surface units rather than treating shared-surface blocks as independent replications. Detailed six-design ranking analyses remain explicitly subset analyses in the supplement.

Estimation and allocation risk

The primary maximum likelihood SAR fit included an intercept, treatment, the true regime-specific spillover covariate and grid incidence. Our concentrated-likelihood implementation matched `spatialreg::lagsarlm` in validation; confidence intervals were $\hat{\tau} \pm 1.96\widehat{\mathrm{SE}}$.[10,11] This assumes the analyst knows the spillover form. A grid sensitivity omitted spillover. Aliases, warnings, failed estimates/intervals and matrix rank were recorded. Rook Checkerboard has $WZ = 1 - Z$, so treatment and spillover are not separately identified; its retained treatment coefficient estimates $\tau - \gamma$. Under queen control-only spillover, identification relies heavily on the grid boundary. The rook estimate does not recover the target effect; queen Checkerboard remains identified but can be imprecise.

For assignment a, allocation risk is $m(a) = E_\epsilon[(\hat{\tau} - \tau)^2 \mid a, X]$. Nested outcome replication estimates its distribution before calculating allocation q90 and the estimated mean of the worst 10% of sampled assignments. These differ from quantiles of realized errors or scenario-level MSE. A focused grid pilot used two fixed surfaces per configuration, queen parameter corners, 100 assignment draws and 100 outcomes per unique assignment, with a selected same-assignment refinement to 400 outcomes. Its findings are supporting pilot evidence, separate from the full grid comparison.

The repeated-outcome step is essential to interpreting downside. For a fixed incidence surface and assignment, random outcome noise can produce a large individual estimation error even when that assignment's expected MSE is modest. We therefore average squared errors over outcomes before ordering assignments by estimated risk. Conversely, a design with only one possible assignment has no between-allocation risk distribution to protect against, but its conditional mean error can still be high. Finite outcome replication adds noise to the estimated risks, so observed tail separation and tail membership require separate scrutiny from the precision of a design's overall mean.

Observed-incidence application

We aggregated corrected county-year SUD counts prepared by Habib[12] and SEER populations aged 18–64 into 58 community-college service-area clusters partitioning North Carolina's 100 counties. The corrected numerator includes heart-failure deaths previously excluded as presumed non-sudden (Habib, personal communication, September 25, 2026): 23,523 deaths over 25,594,321 person-years during 2018–2021. Annual rates were deaths/population per 100,000. Every county mapped to one cluster; annual cluster totals reproduced statewide totals. Observed rates were held fixed with no synthetic incidence generation (Figure 1). The four years were separate planning inputs, not longitudinal intervention follow-up.

Table 1. Allocation rules on the 100-cluster grid and their North Carolina adaptations. Incidence means the simulated grid covariate or observed annual cluster rate, respectively. Detailed rounding, tie-breaking and partition rules are in the supplementary material.

Design	Allocation rule
Checkerboard	Alternate grid neighbors; deterministic greedy graph interspersion in NC.
High Incidence Focus	Treat the highest-incidence half, randomizing ties.
Saturation Quadrants	Randomly order 20/40/60/80% saturation within four grid quadrants; four frozen geographic regions in NC.
Isolation Buffer	Greedy random maximal independent set: treated clusters cannot be neighbors.
2×2 Blocking	Treat two of four within grid blocks; randomize rounded half-budgets in frozen NC spatial/incidence/population groups.
Balanced Quartiles	Randomize half within each incidence-rank quartile, with exact overall half-budget.
Balanced Halves	Randomize half within each incidence-rank half, with exact overall half-budget.
Incidence-Guided Saturation	Order 80/60/40/20% by quadrant mean incidence; by regional mean annual cluster rank in NC.
SRS	Completely randomize half of clusters (50 grid; 29 NC).

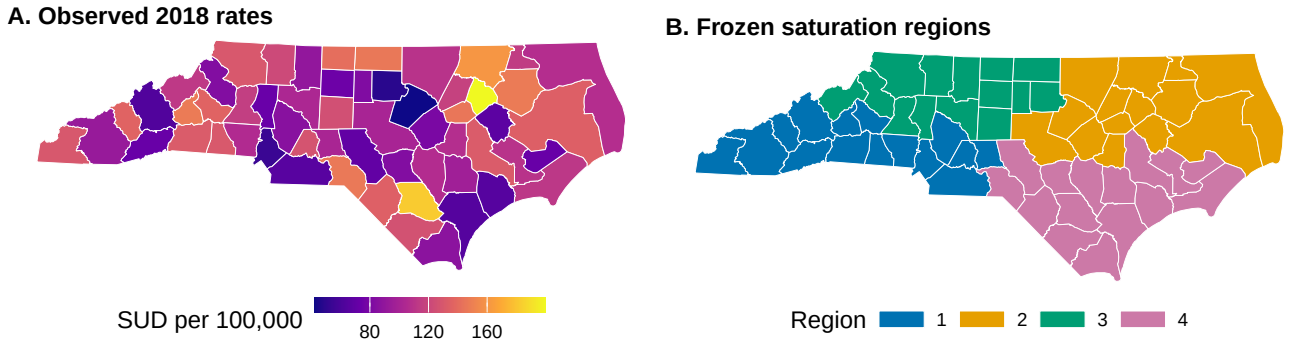


Figure 1. Observed North Carolina planning inputs. (A) Illustrative 2018 SUD rates per 100,000 population aged 18–64 for the 58 service-area clusters; all four years are evaluated separately. (B) The frozen four-region partition used by both saturation strategies in every year. Incidence guides primary allocation only; it does not enter the education-outcome baseline. All-year maps and assignment examples are in the supplementary material.

Primary education outcomes used Equation 1 with $\beta = 0$, $\tau = 1$ and the named cached 58-cluster W. Incidence directed allocation only. The fit used intercept, treatment and true spillover without incidence adjustment. Thus the application does not assume historical SUD incidence predicts education performance or modifies the constant intervention effect.

All nine strategies were adapted before performance evaluation. Checkerboard became a deterministic greedy graph-interspersion assignment, not a literal checkerboard. Saturation used four frozen geographic regions selected from 250 k-means candidates by compactness and population/cluster/county balance, using pooled person-years/4 as reference population. Guided saturation ranked annual per-capita cluster rates, averaged cluster ranks equally within regions, then ordered regional saturations 80/60/40/20%. The grid rule averages its incidence covariate directly. The application mean-rank rule aligns with rank-scaled Poisson surfaces but differs from continuous grid incidence; it also differs from pooling deaths/population. Spatial blocking used 14 frozen yearly k-means groups based on standardized location, incidence ranks and log-population. Other strategies retained their mechanisms. SRS and incidence-balancing/targeting treated 29 clusters; Balanced Halves randomly assigned 14/15 within its two halves. Rounded local budgets were retained and reported rather than equalized retrospectively.

These adaptations preserve each strategy's allocation mechanism while acknowledging the irregular graph and unequal populations. Geographic regions replace rectangular quadrants; frozen groups replace uniform four-cluster blocks. Neither is an exact copy of the grid design. We selected partitions without examining simulated estimation performance and retained their rounding rules, including resulting treatment-count differences. Cluster counts and working-age population shares accompany accuracy because treating half the clusters does not imply reaching half the population. The population denominator used to calculate historical rates is also distinct from the eventual sample of healthcare professionals whose education outcomes would be measured.

Primary queen settings crossed $\rho = 0, 0.2, 0.5$, $\gamma = 0.5, 0.8$, both regimes, four years and nine designs. Sensitivities added a hypothetical rank-scaled incidence baseline together with incidence adjustment, used rook parameter corners, or replaced guided saturation's regional mean rank with mean raw rates or pooled regional rates. The total was 1,248 reporting rows from 936 computational sources. Identical supports reused computation with explicit source IDs; annual budgets and population shares were

recalculated. Shared annual sources are not independent confirmations.

We drew 100 assignments and 100 independent outcomes per unique assignment, using 1,000 outcomes for proven singleton supports. Prespecified adaptive replication targeted relative MCSE(mean MSE)$\leq 5\%$ and MCSE(coverage)≤ 0.01; duplicate-frequency covariance and both allocation/outcome variation were accounted for. A separate queen-corner extension retained assignments and raised stochastic outcome replication to 400. Opposite outcome halves selected and evaluated worst-decile assignments. Key-based seeds and manifest checks protected reproducibility and refused stale checkpoints. Further details, matched sensitivity tables and failure diagnostics are in the supplement.

Results

Grid accuracy and pilot downside

Table 2 reports the full queen benchmark at $\tau = 1$; Figure 2 separates incidence configurations in the detailed six-design subset. Figure 3 shows the error decomposition for one prespecified Poisson configuration; variance dominates its queen MSE in both regimes. Guided saturation's pooled mean MSE was 0.0907 against SRS 0.1285, but regime separation was essential: control-only MSE was 0.1355 against 0.2146 (ratio 0.631), whereas both-arms MSE was 0.0459 against 0.0424 (ratio 1.083). Plain saturation had similar accuracy; an extra benefit from incidence guidance was not established. SRS and incidence balancing were competitive under both arms. Identified designs' coverage was approximately 94%, below 95%; rook Checkerboard was nonidentified and substantially biased. Rook results and effect-size/configuration breakdowns are reported separately in the supplement.

The regime difference was especially large for interspersed assignments. Checkerboard MSE increased from 0.0624 under both arms to 2.1114 control-only; blocking increased from 0.0623 to 0.5761. Their poorer control-only accuracy reflects imprecision rather than a comparable increase in bias. In the selected Poisson configuration, empirical variance accounts for more than 97% of mean MSE for every displayed design and regime (Figure 3). Buffer behaves differently: no treated cluster has a treated neighbor, so multiplying its neighboring-treatment exposure by the control indicator leaves that exposure unchanged. Its two spillover regimes therefore define the same outcome model; small observed mean differences arise from separate Monte Carlo draws.

The detailed six-design queen pooled MSE ordering was stable across the five incidence configurations and five treatment-effect sizes. This does not imply a fixed conditional ranking. Under control-only spillover, buffer and guided saturation were nearly level at $\tau = 2$ (MSE 0.1596 and 0.1604); at $\tau = 3$, buffer was lower (0.1596 versus 0.1822). Guided saturation remained lowest in pooled MSE within that subset, but the crossover limits a blanket claim across effect sizes. Effect-size comparisons also reused assignments and noise, so they are sensitivity comparisons rather than independent confirmations. Full parameter/configuration and rook breakdowns in the supplement retain these distinctions.

The grid sensitivity omitting spillover illustrates why an apparently favorable error metric is insufficient. Among the six-design comparisons at $\tau = 1$, excluding nonidentified rook Checkerboard, pooled bias ranged from approximately -0.12 to -0.40 and coverage from 0.51 to 0.90. Some designs had lower MSE under this misspecified fit because their reduced variance outweighed increased bias. That trade-off did not restore valid treatment-effect inference. These sensitivity results concern omission of spillover; they do not establish robustness to an incorrectly specified exposure network or spillover functional form.

The selected-surface grid pilot supported saturation's control-only downside pattern. Guided saturation's estimated worst-decile conditional MSE was 0.1936 against SRS 0.3486; under both arms it was 0.0582 against 0.0544. Outcome refinement preserved the control-only direction but did not remove uncertainty about allocation tails. These are pilot values and do not replace full-study estimates.

Completed application

All 1,248 application reporting settings completed and passed precision gates. Production used 7,598,000 independent outcome fits; the separate extension added 2,280,000, for 9,878,000 distinct outcomes after counting copied prefixes once. Maximum relative MCSE(mean MSE) was 0.04871 and maximum MCSE(coverage) 0.00892. There were no warnings, aliases, boundary fits or estimation failures. All main stochastic sources passed at 100 assignments and 100 outcomes; two singleton baseline-sensitivity sources required 2,000 outcomes.

Under control-only spillover, plain saturation's yearly descriptive mean MSE was 0.22329 against SRS 0.30761, a 27.4% reduction. Guided saturation's reductions were 21.1–29.9% depending on year (Figure 4; absolute annual values in the supplementary material). Both saturation designs had lower MSE than SRS in every primary parameter setting, including setting-specific approximate MC intervals. Neither was uniquely best. High Incidence Focus's control-only accuracy varied markedly across years. Under both arms, SRS mean MSE was 0.07638, plain saturation 0.07518 and guided saturation 0.07428–0.07682. These small differences do not establish equivalence or universal optimality. Primary yearly mean coverage was approximately 92.6–93.6%, below nominal; passing precision targets did not repair undercoverage.

The annual comparison preserves the broad regime distinction while showing changes introduced by the NC geography. Graph Checkerboard is identified, yet has roughly three times SRS MSE in both regimes. Buffer is close to SRS control-only but has approximately 3.74 times its both-arms MSE. Spatial blocking is close to SRS under both arms and more variable across planning years control-only (Figure 4). Thus transferring a grid mechanism does not transfer its numerical ranking unchanged. These year-specific descriptive averages are conditional on fixed observed incidence and the adapted supports; they neither incorporate uncertainty in historical rates nor independently replicate the incidence-independent assignments reused across years.

Table 2. Full grid benchmark at $\tau = 1$, queen contiguity. Each regime-specific entry averages 80 scenarios (five incidence configurations, four ρ and four γ values). MSE and coverage use the oracle fit. SRS is complete randomization of 50 clusters. No value pools queen and rook.

	Both arms		Control only	
Design	MSE	Coverage	MSE	Coverage
Checkerboard	0.0624	0.940	2.1114	0.942
High Incidence Focus	0.1617	0.941	0.3292	0.935
Saturation Quadrants	0.0442	0.942	0.1359	0.929
Isolation Buffer	0.1608	0.939	0.1595	0.939
2×2 Blocking	0.0623	0.937	0.5761	0.941
Balanced Quartiles	0.0421	0.940	0.2195	0.932
Balanced Halves	0.0420	0.939	0.2179	0.933
Incidence-Guided Saturation	0.0459	0.939	0.1355	0.934
SRS	0.0424	0.941	0.2146	0.936

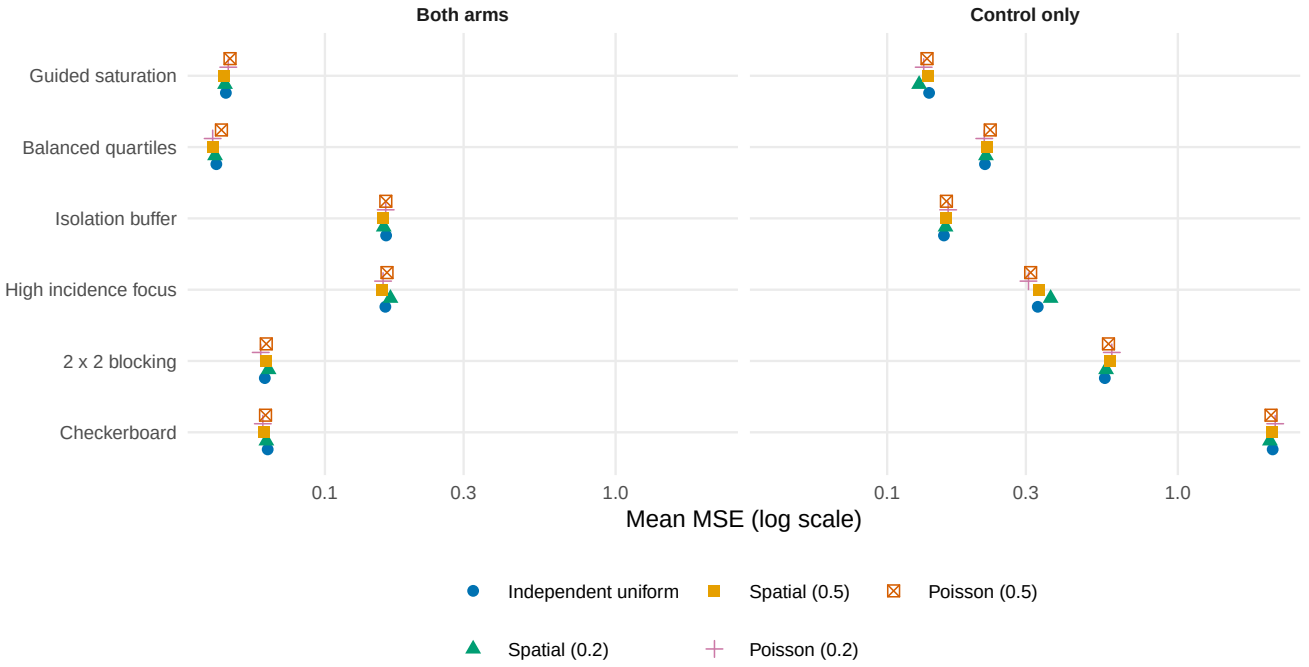


Figure 2. Grid mean MSE at $\tau = 1$ with queen weights, for the detailed six-design subset. Points keep the five incidence configurations separate, averaging 16 outcome-dependence/spillover-magnitude settings within each configuration and regime. Parentheses in the legend give incidence dependence ρ_X. Both panels use the same logarithmic scale. Table 2 includes all nine designs. Parameter-specific displays remain in the supplementary material; no pooled Monte Carlo intervals are implied.

At the refined queen corners with $\gamma = 0.8$, control-only plain saturation had estimated q90 31.8–32.9% below SRS and worst-decile mean 34.6–37.8% below; guided saturation reductions were 20.8–38.6% and 20.7–43.4%. Cross-selected outcome halves retained this pattern (detailed tail displays in the supplementary material). However, only ten sampled draws formed the worst decile. All 112 non-singleton reporting rows flagged uncertain tail-boundary membership. Both-arms guided saturation had half-sample Jaccard overlap 0–0.25, with about 61–86% of raw estimated-risk variance attributable to outcome MC noise. Small both-arms tail differences cannot support superior downside protection. Singleton Checkerboard and High Incidence Focus have no allocation variation for these inputs, but can still have substantial estimation error. Across all nine strategies (supplementary material), Isolation Buffer also had a lower control-only worst-decile mean than SRS (0.62 of SRS in the grid pilot and 0.91 in the application) but 3.7–3.8 times SRS mean MSE under both arms. High Incidence Focus's fixed application allocation was below the SRS worst-decile mean in both regimes (0.64 control-only, 0.76 both arms), but its grid-pilot both-arms mean was 3.91 times that of SRS.

Budgets differed (annual cluster/population tables and displays in the supplementary material). Graph Checkerboard treated 27 clusters under queen; buffer averaged 15.89; spatial blocking treated 28/31/27/29 by year, including a singleton group always assigned control under the retained rounding rule. High Incidence Focus treated 29 clusters but approximately 22–24% of working-age population; buffer reached about 27%. These shares describe geographic reach, not healthcare-professional enrollment or cost. Graph Checkerboard was identified but imprecise, unlike the exact rectangular-rook alias.

The hypothetical baseline/X-adjusted sensitivity increased High Incidence Focus MSE roughly 3.68–4.44-fold under

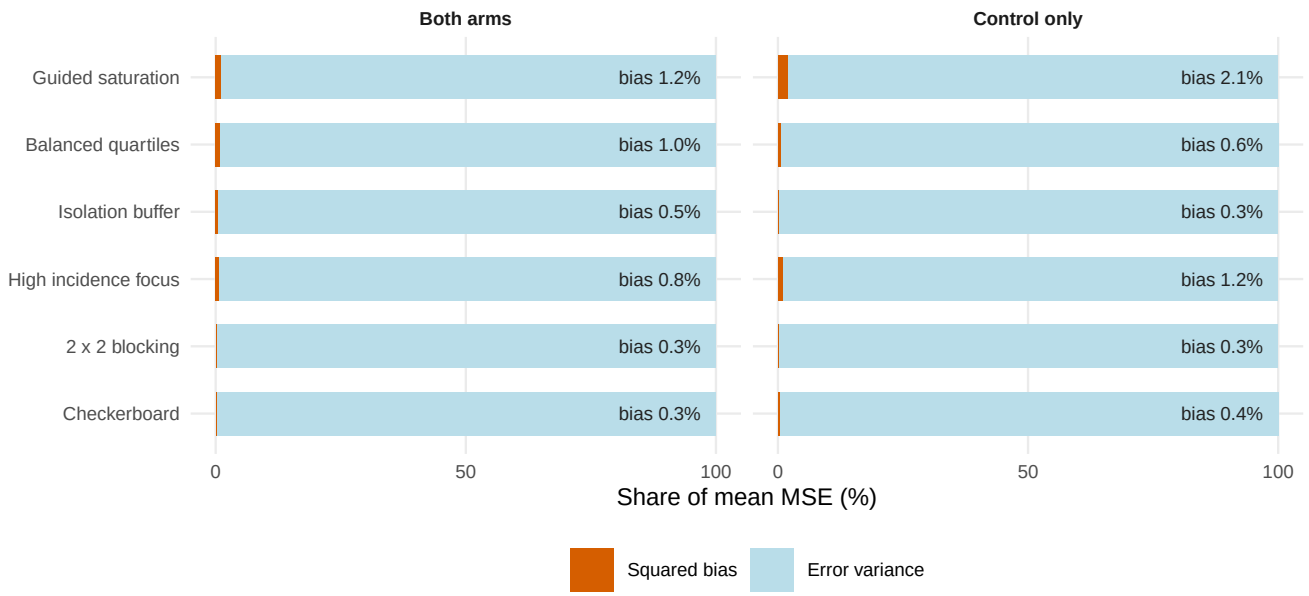


Figure 3. Composition of grid mean MSE for the six-design subset, Poisson incidence with $\rho_X = 0.2$, $\tau = 1$ and queen weights. Squared bias and empirical error variance are averaged over 16 settings separately in each spillover regime, then divided by mean MSE. Labels show the squared-bias percentage. Empirical variance uses $(249/250)\mathrm{SD}^2$, so its sum with squared bias equals reported MSE exactly. This figure describes one incidence configuration, not all grid settings.



Figure 4. Completed primary application: yearly mean MSE divided by year- and regime-matched SRS MSE, for all nine designs with queen weights and education outcomes. Each cell averages six ρ/γ settings equally before taking the ratio. Blue indicates lower error than SRS, orange higher error; colors use a logarithmic ratio scale and cells print the ratios. SRS mean MSE is 0.07638 under both arms and 0.30761 control-only. Shared annual sources are not independent replications. Absolute MSE, bias, coverage, budgets and setting-specific comparisons are in the supplementary material.

both arms and 1.42–2.52-fold control-only. Treatment–incidence alignment reduced information after adjustment; baseline and adjustment changed jointly. Matched rook/queen corners preserved broad conclusions. Alternative regional summaries changed some region orders, with no retrospective replacement of the prespecified mean-rank rule.

Discussion

The completed grid and observed-incidence application support conditional guidance: consider saturation when spillover primarily reaches controls, and retain SRS among competitive options when both arms receive exposure. Average MSE and allocation downside should be evaluated together. A restricted design's near-zero allocation variability

is insufficient if its conditional error is high. Conversely, small estimated tail differences with noisy membership are insufficient to dismiss SRS. Isolation Buffer's control-only downside protection is useful only when both-arms spillover is implausible and fewer treated clusters are acceptable.

Incidence guidance did not establish an accuracy advantage over plain saturation. Historical incidence can still express implementation priorities, but that purpose should be distinguished from improved estimation. High Incidence Focus illustrates the trade-off: low conditional both-arms error in the primary education model came from a fixed assignment, not a demonstration of randomized causal identification. Its sensitivity to incidence adjustment also prevents a general recommendation based on that result alone.

For trial planning, the combined findings favor evaluating plausible exposure regimes before selecting an allocation. Saturation creates variation in local treatment proportions while retaining treatment and control clusters; its favorable control-only accuracy is consistent with the need to distinguish direct treatment from neighboring exposure. Under both-arms spillover, SRS and incidence-balanced assignments already provide competitive estimation, making a modest mean difference an insufficient reason to impose a more restricted design. Incidence-targeted assignment can still serve a substantive delivery objective, but a fixed high-incidence allocation needs its own justification and cannot inherit the randomization-based interpretation of SRS.

These comparisons complement spatial analysis rather than replace it. The primary fit knows the spillover mechanism; the grid sensitivity omitting it showed bias and poor coverage. We did not establish performance with incorrect exposure networks, heterogeneous treatment effects or participant-level sampling error. The grid and one irregular geography do not span all trial structures, and unequal treatment budgets prevent interpreting the rankings as equal-cost optimization. Historical incidence was fixed; measurement uncertainty and change between planning and implementation were not propagated.

Wald coverage remained below nominal even when Monte Carlo precision was adequate. Interval calibration must be addressed before adopting this analysis for a final protocol. Allocation-tail estimates are conditional on sampled masks and finite outcome replication; the extension improved risk estimation for the same masks rather than providing fresh allocation confirmation. Shared annual supports also preclude treating repeated performance as independent evidence.

A practical decision should also compare recruitment and exposure consequences alongside MSE. Buffer's smaller treated-cluster budget and High Incidence Focus's smaller population reach mean that these results do not answer which design is most efficient for a fixed implementation cost or participant sample. Before using them to choose one pilot allocation, investigators would need to define the education endpoint, identify eligible professionals, specify cluster and participant budgets, and assess whether geographic contiguity represents actual communication. Those choices could alter outcome precision or the exposure network. Interval calibration is another protocol-stage requirement: precise simulation estimates of subnominal coverage identify a limitation, rather than validate the reported Wald intervals.

The education application is a completed design evaluation, not an implemented intervention or evidence that education reduces SUD. Final decisions require a specified education endpoint, eligible professionals, budget and plausible communication network. Within the evaluated assumptions, saturation offers favorable average and descriptive downside performance under control-only spillover; SRS remains a reasonable comparator and candidate under both-arms exposure.

Author contributions

AW developed the method, designed the simulation study, implemented the application study, prepared all figures and wrote the manuscript. AH prepared the North Carolina SUD datasets for the application study. RS proposed the SUD prevention education pilot. FL provided methodological guidance and review.

Declaration of conflicting interests

The authors declare that there is no conflict of interest.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability

All data supporting the findings of the simulation study were generated by simulation. The simulation data and R code are available from the authors on request, with the authors' permission. The North Carolina death-certificate data used in the application are restricted and are not publicly available.

Supplementary material

The supplementary document supplies exact design rules, parameter grids, estimation diagnostics, detailed grid results, application maps, annual performance/budgets and matched sensitivities.

References

1. Donner A and Klar N. *Design and analysis of cluster randomization trials in health research*. Arnold, 2000. ISBN 978-0-470-71100-2 978-0-340-69153-3.
2. Hayes RJ and Moulton LH. *Cluster Randomised Trials, Second Edition*. Chapman and Hall/CRC, 2017. ISBN 978-1-315-37028-6. DOI:10.4324/9781315370286. URL https://www.taylorfrancis.com/books/9781315370286.
3. Hudgens MG and Halloran ME. Toward Causal Inference With Interference. *Journal of the American Statistical Association* 2008; 103(482): 832–842. DOI:10.1198/016214508000000292. URL https://www.tandfonline.com/doi/full/10.1198/016214508000000292.

4. Jarvis C, Di Tanna GL, Lewis D et al. Spatial analysis of cluster randomised trials: a systematic review of analysis methods. *Emerging Themes in Epidemiology* 2017; 14(1): 12. DOI:10.1186/s12982-017-0066-2. URL https://ete-online.biomedcentral.com/articles/10.1186/s12982-017-0066-2.
5. Crisp AM, Halloran ME, Longini IM et al. Covariate-constrained randomization with cluster selection and substitution. *Clinical Trials* 2023; 20(3): 284–292. DOI:10.1177/17407745231160556. URL https://journals.sagepub.com/doi/10.1177/17407745231160556.
6. Baird S, Bohren JA, McIntosh C et al. Optimal design of experiments in the presence of interference. *The Review of Economics and Statistics* 2018; 100(5): 844–860. DOI:10.1162/rest_a_00716.
7. Cai C, Zhang X and Airoldi EM. Independent-set design of experiments for estimating treatment and spillover effects under network interference. In *International Conference on Learning Representations*. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/a204aa68ab4e970e1ceccfb5b5cdc5e4-Abstract-Conference.html. Published ICLR 2024 version of arXiv:2312.04026; key kept for existing citations.
8. Leung MP. Rate-optimal cluster-randomized designs for spatial interference. *The Annals of Statistics* 2022; 50(5): 3064–3087. DOI:10.1214/22-AOS2224.
9. Walther A, Van Deinse T and Lin FC. Spatial cluster randomized trials - sampling design with spillover effects & spatial dependence. *BMC Medical Research Methodology* 2026; DOI:10.1186/s12874-026-02968-0. URL https://link.springer.com/article/10.1186/s12874-026-02968-0. Published online 26 August 2026.
10. Ord K. Estimation Methods for Models of Spatial Interaction. *Journal of the American Statistical Association* 1975; 70(349): 120–126. DOI:10.1080/01621459.1975.10480272. URL http://www.tandfonline.com/doi/abs/10.1080/01621459.1975.10480272.
11. Bivand R and Piras G. Comparing Implementations of Estimation Methods for Spatial Econometrics. *Journal of Statistical Software* 2015; 63(18). DOI:10.18637/jss.v063.i18. URL http://www.jstatsoft.org/v63/i18/.
12. Habib A. *Temporal Trends and Social Determinants of Sudden Death Mortality in North Carolina*. Master's paper, Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC, 2026. Unpublished.